

Vision-Based Vehicle Tracking using Particle Filtering



AA273, Winter 2026

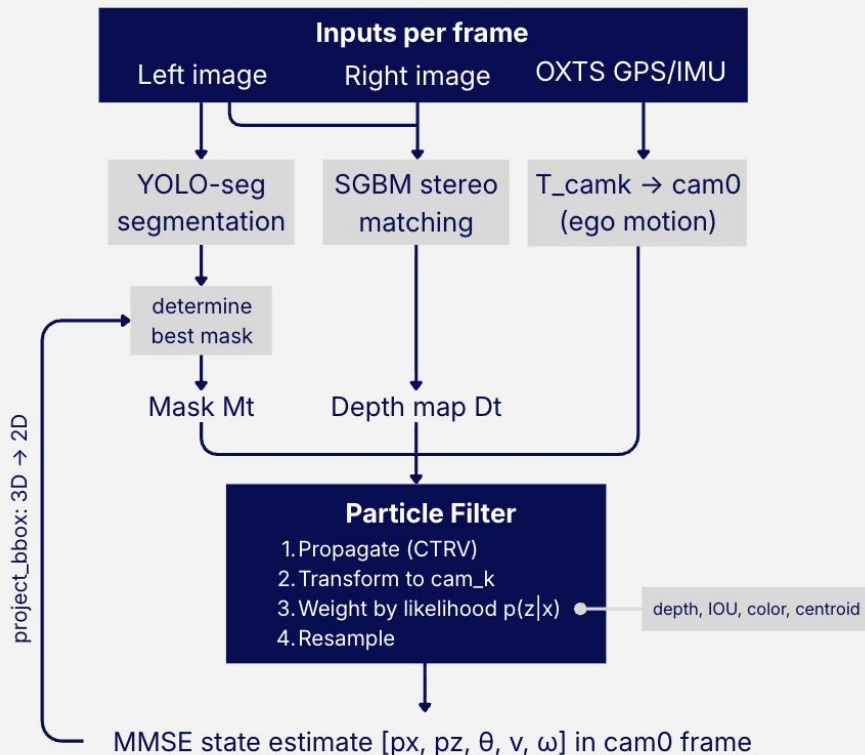
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Problem & Goal

- Track a vehicle's 5D state over time via a sequence of image frames. Position and heading are expressed in a fixed reference frame anchored at frame 0 (cam0).
- Data: driving sequence using only stereo camera images and onboard GPS/IMU data (of camera vehicle, not vehicles being tracked)
- Ground truth 3D positions and headings are provided by the KITTI tracking benchmark for evaluation



Pipeline Overview



3D $\rightarrow$ 2D projection: bridge between particle state space and image observations



Dynamics Model

- Assumes car moves along circular arcs
- Speed noise and turn rate noise injected each step to model changes in speed and heading

CTRV kinematic model:

State vector:

$$x_t = \begin{bmatrix} p_x \\ p_z \\ \theta \\ v \\ \omega \end{bmatrix}$$

Process noise:

$$\eta_v \sim \mathcal{N}(0, \sigma_v^2), \quad \eta_\omega \sim \mathcal{N}(0, \sigma_\omega^2)$$

Dynamics equations:

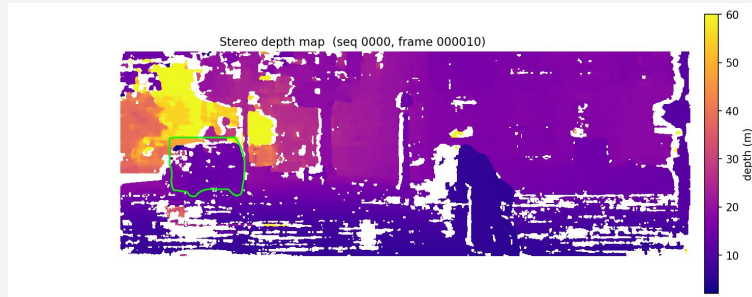
- Straight line: $\begin{bmatrix} p_x \\ p_z \end{bmatrix}_{t+1} = \begin{bmatrix} p_x + v \cos \theta \Delta t \\ p_z - v \sin \theta \Delta t \end{bmatrix}$
- Turning: $\begin{bmatrix} p_x \\ p_z \\ \theta \\ v \\ \omega \end{bmatrix}_{t+1} = \begin{bmatrix} p_x + \frac{v}{\omega} (\sin(\theta + \omega \Delta t) - \sin \theta) \\ p_z + \frac{v}{\omega} (\cos \theta - \cos(\theta + \omega \Delta t)) \\ \theta + \omega \Delta t \\ v + \eta_v \\ \omega + \eta_\omega \end{bmatrix}$

Measurements & Likelihoods

YOLO26-seg Segmentation Mask M_t



OpenCV SGBM Stereo Depth Map D_t



Depth Likelihood: 10th percentile depth of masked depth map vs depth of closest 3D bbox point of particle.

Mask IOU Likelihood: IOU of M_t with convex hull mask of projected 3D bbox of particle.

Color Likelihood: Bhattacharyya distance between color histogram of masked image and particle masked image (using convex hull mask of projected bbox) in HSV space (since H and S illumination invariant).

Centroid Likelihood: Centroid of M_t vs centroid of convex hull mask of projected bbox of particle.

Likelihood Model

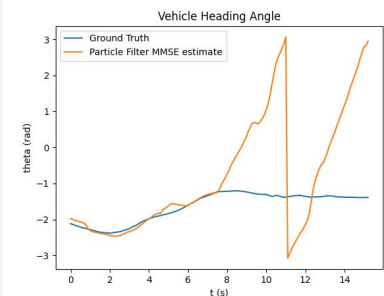
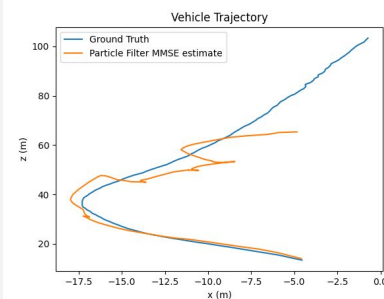
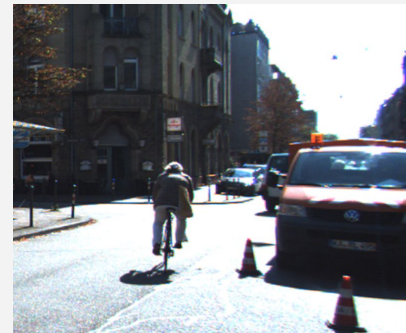
4 likelihood terms (log-additive)

Depth	$p(D_t \mid x_i) = \exp\left(-\frac{(Z_{\text{pred}} - Z_{\text{meas}})^2}{2\sigma_Z^2}\right)$
Mask IOU	Projected particle mask IOU $p_{\text{IOU}}(M_t \mid x_i) = \exp(-\alpha (1 - \text{IoU}(\hat{M}_i, M_t)))$
Color	HSV histogram similarity $p_{\text{color}}(M_t \mid x_i) = \exp(c_H \cdot D_B(H_{\text{ref}}, H_i))$
Centroid	Gaussian on projected centroid distance $p_{\text{cen}}(M_t \mid x_i) = \exp\left(-\frac{\ \hat{c}_i - c_t\ ^2}{D \cdot \sigma_c^2}\right)$

Combined weight update: $\log w_i = \log p_{\text{depth}} + \log p_{\text{IOU}} + \log p_{\text{color}} + \log p_{\text{cen}}$

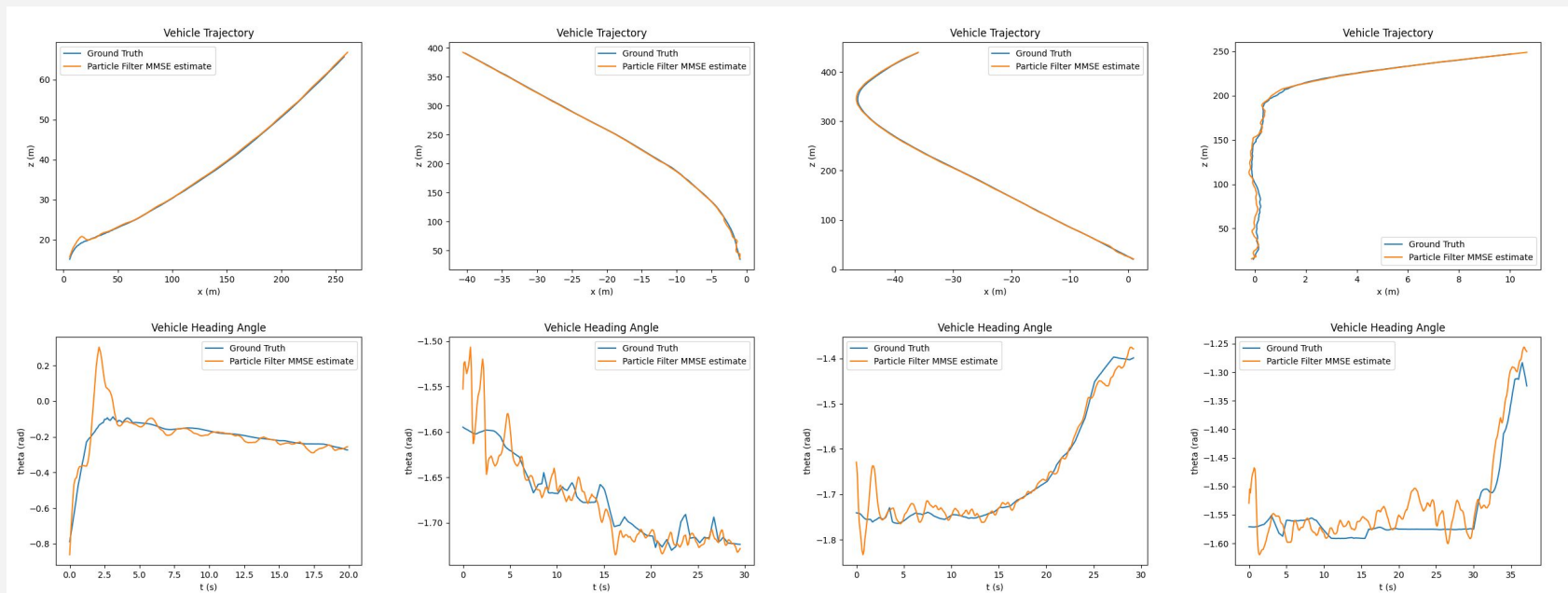
Challenges

- Particle filter behaved quite well when measurements were reasonable.
- Performance degraded when measurements were poor/incorrect.
 - Stereo depth measurements around vehicle were consistently accurate when vehicles had sufficient features for stereo matching
 - YOLO vehicle mask measurements suffered in the following cases, often resulting in vehicle swapping:
 - Significant glare (affecting color detection, usually only an issue with other vehicles nearby)
 - Vehicle occlusion (severe partial and total)
 - Similar vehicles (in color and location)
- Potential Solutions:
 - Methods for determining when measurements are poor, then falling back on dynamics temporarily.
 - More sophisticated mask identification (learned approach as opposed to classical, which is used currently).



Results

- When the correct mask could be identified by YOLO, the filter tracked the desired vehicle quite well.



Results

- The metrics used to assess filter performance were the x, y, planar, and heading RMSE. The values below are for the previous plots.

Sequence	x RMSE (m)	z RMSE (m)	Planar RMSE (m)	Heading RMSE (rad)
0004	1.853	0.726	1.990	0.0822
0005	0.202	1.722	1.734	0.0219
0010	0.203	0.840	0.865	0.0258
0011	0.134	0.797	0.808	0.0348

Conclusions and Future Work

Conclusions:

- When mask measurements are reasonable and detect the correct vehicle mask, the filter is able to track quite well in both position and heading.
- Incorrect mask selection/detection is often caused by environmental challenges such as glare, occlusion, and the presence and visual similarity of nearby vehicles.
- The chosen likelihood models complement each other and are able to robustly filter particles to track the desired state.

Future Work

- More advanced approaches could be considered for robust mask selection (such as learning based).
- The KITTI dataset mostly includes trajectories with fewer turns and stops. To fully evaluate the robustness of the tracker, it would be beneficial to test on sequences with greater variety of vehicle movement.
- Metrics for determining when mask and/or depth measurements are poor enough to fall back strictly on transition dynamics.

References

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